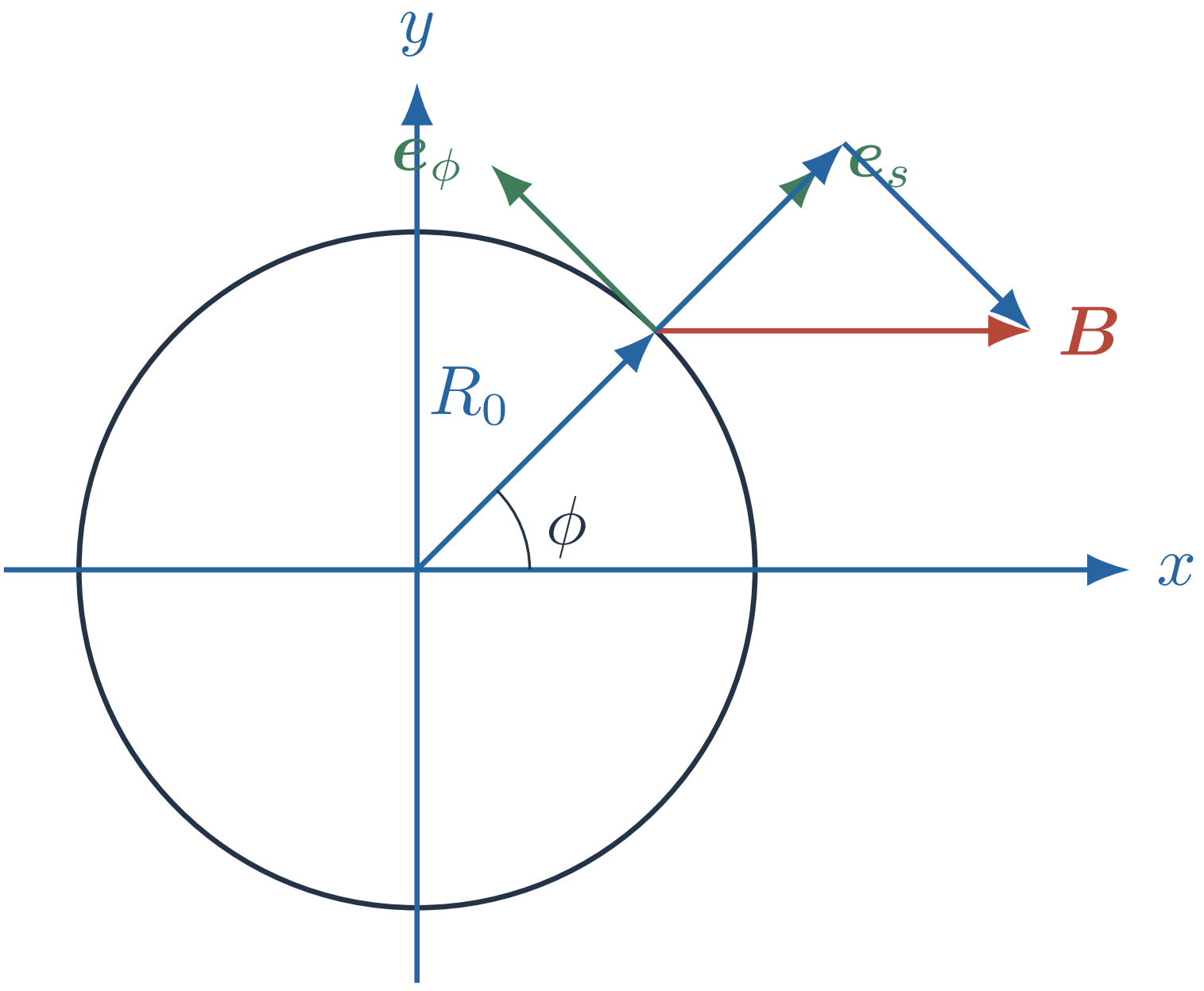

$$\boldsymbol{B} = B\cos\phi\,\boldsymbol{e}_s - B\sin\phi\,\boldsymbol{e}_\phi$$